Lab Experiment 6

Aim :

To implement logical instruction LR, RS, Arithmetic right shift, Arithmetic left shift, RR, RL, RR with carry and Rl With carry. Implement the logic using 8 bit shift register

Theory

Shift micro-operations move bits left or right within a register and are used in tasks like data storage, serial transmission, multiplication, division, and logical operations. They are often combined with arithmetic and logic operations for efficient processing.

A logical shift micro-operation involves moving the bits of data in a register either to the left or right, with zeroes being introduced into the empty positions. This operation is used primarily in binary arithmetic and data processing tasks.

There are two main types of logical shift micro-operations:

Logical Left: In this shift, each bit is moved to the left by one position. The LSB is filled with zero and the MSB is rejected, Used in multiplication of unsigned binary numbers.

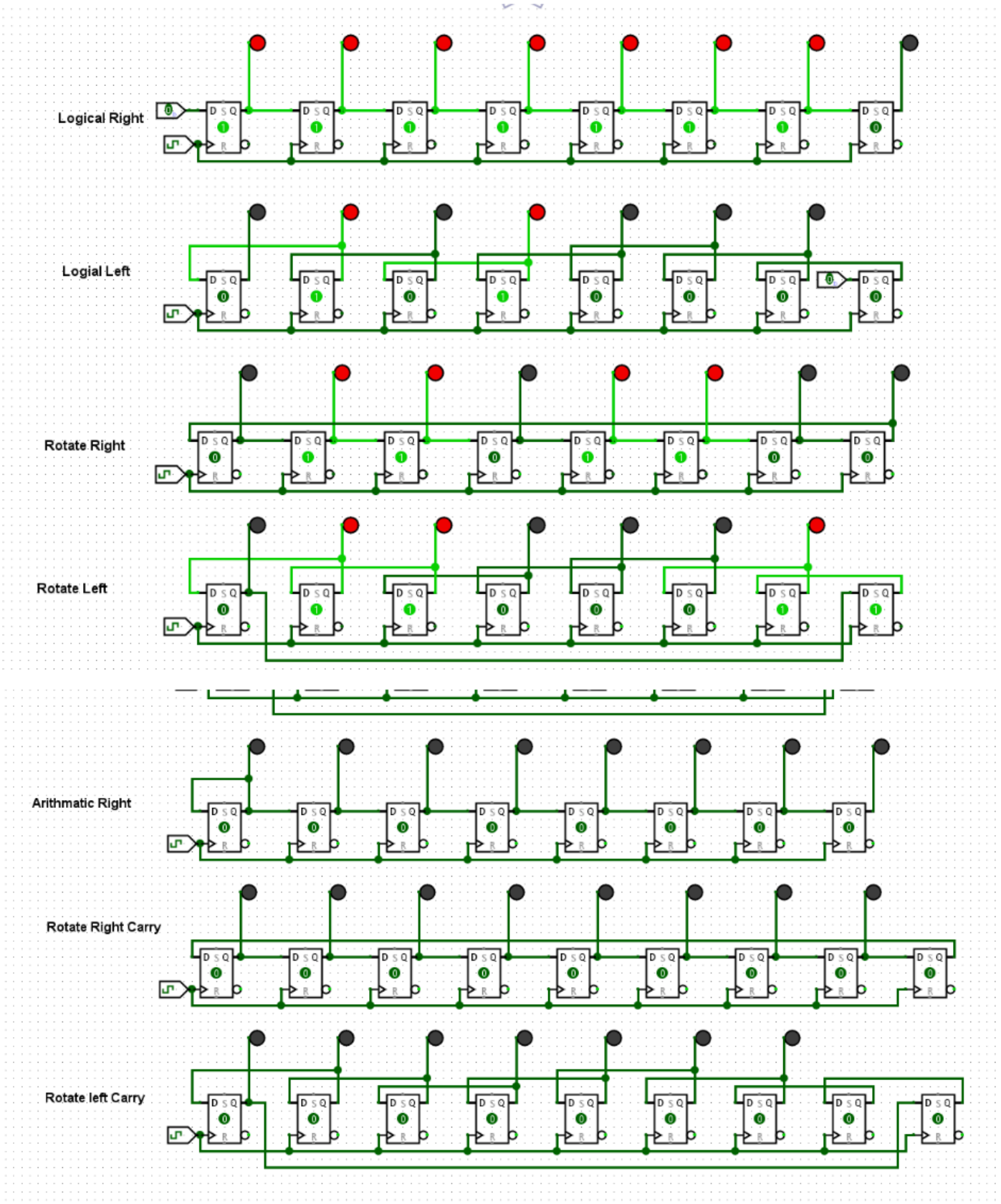
Logical Right: In this shift, each bit moves to the right one by one. LSB is discarded and the MSB is filled with zero ,Used in division of unsigned binary numbers.

The arithmetic shift micro-operation moves the signed binary number to the right position. In this shift, each bit is moved to the right one by one and the LSB bit is rejected and the MSB is filled with the value of the previous MSB ,Used for dividing signed binary numbers by powers of 2. Arithmetic left shift also exists but it works the same way as logical left.

The circular shift or Rotate shift circulates the bits in the sequence of the register around both ends without any loss of information.

Following are the two ways to perform the circular shift:
Rotate Left: In this micro shift operation each bit in the register is shifted to the left one by one. After shifting, the LSB becomes empty, so the value of the MSB is filled in there.
Rotate Right: In this micro shift operation each bit in the register is shifted to the right one by one. After shifting, the MSB becomes empty, so the value of the LSB is filled in there.
The rotate Shifts can also be made by adding a carry at the end of the register which connects
to the MSB while getting its value from the LSB

Procedure :



Lab Implementation

Result :

Successfully implemented shift instructions using a register made in a previous experiment.